

3 Methods

This section contains descriptions of the sampling procedure and details for the DC experts, LLMs, and DC solutions. Also, the EE questions, experimental setup, and simulation. Further, the section describes the chosen tests and analytical procedures for all experiments.

3.1 Sampling

Decisions on the number of experts for the EE are sometimes determined by the desired expert mix, availability and willingness, as well as budget and time constraints [32, 40]. Among the studies that discuss the number of experts per study are Knol and Colleagues (2010), who propose an average of 6 per study [25]. Several publications recruited more than 6 participants, suggesting availability, and study objectives determine the number of experts [18, 32, 62, 68]. Hence, in our initial study [3], we left it open and recruited experts based on availability.

This section elaborates on the sample and sampling procedure of the DC solutions, DC experts, and the LLMs. The studied DC solutions are part of the solutions proposed in the *Anonymized* project, which also provided ethical clearance for the EE with DC experts. The project focuses on introducing innovative approaches to power system implementation and on integrating DC solutions into four key demonstration areas: ports, buildings, industry, and data centers [38].

3.1.1 Sampling the DC experts and LLMs. The EE with DC involved 24 experts in two EE sessions held between September and October, 2024. The first session involved 11 experts from within the *Anonymized* project consortium. The other session involved 13 experts referred by colleagues in the DC industry. The distribution of expertise of experts from both studies is presented in Table 1. Email was used for initial communication and sharing of the consent forms with experts prior to performing the elicitation tasks.

Sampled LLMs as presented in Table 2, we used 4 general-purpose open-source LLMs to contribute to the open-source community. To reduce computational requirements, the selected models have quantized model files below 10GB and are trained with 7B to 14B parameter counts as distributed via Ollama. The models are Gemma3:12B (Gemma3:12B), LLaMa-Pro (LLaMa-Pro), Mistral (Mistral), and Phi4 (Phi4). For each model, we generated 6 variants, 2 for each role type (i.e., generalist (2), normative (2), and subject-matter (2)), at temperatures ranging from 0.65 to 0.90, with a step size of 0.05.

Table 1. The DC Experts Sample

Expertise	Count
DC power electronics	11
Academia / society	4
Ports electrification	3
DC protection systems	2
Data-center electrification	1
Buildings electrification	1
DC interoperability	1
AC/DC power systems	1

3.1.2 The DC solution. A total of 11 out of 30 DC solutions proposed in the *Anonymized* project were used in EE with DC experts and LLMs. The selection criterion for the solutions is their applicability across multiple demonstrators (i.e., demo cross-cutting aspect). The DC solutions are: Smart and sustainable DC cables, DC connectors, static protection

Table 2. The LLMs Information

Model	Quantized model file	Parameter count
Phi-4	9.1 GB	14.7B
Gemma3:12B	8.1 GB	12.2B
LlaMa-Pro	4.7 GB	8.36B
Mistral	4.1 GB	7.25B

systems, semiconductor-based circuit breakers, protection DC system design tool, DC-DC converters, LVAC-LVDC interlink converters, DC measurement devices, DC solution design tools, network design tool for DC solutions, and solid-state circuit breakers. Detailed explanations on each are found in [3].

3.2 The EE questions

The EE questions used and presented in this study, both for DC experts and LLMs, are part of the EE protocol detailed in our earlier publication [3]. The EE questions were maintained between sessions with DC experts and those with LLMs to ensure comparability between them. The questions are feasibility, importance, and barrier selection presented in Table 3. Each participant (both DC experts and the LLMs) answered each question based on the EE setup. For feasibility, the focus was on *overall feasibility*. As for the importance, each participant rated each of the 11 DC solutions. As for the barrier selection, each presented a maximum of 5 barriers from a list of 11.

Table 3. Elicitation Protocol Questions

SNo	Question	Options/Instructions
1.	How feasible is the use of DC in the previously described demonstrators? (overall ratings)	Not able to respond, not feasible, somewhat feasible, feasible, very feasible.
2.	How important is this solution? (For each DC solution)	Not able to respond, not important, somewhat important, important, very important.
3.	Barriers to adoption (a total of eleven barriers were presented for experts to select all that they perceive apply)	Select the top 5.

3.3 The experimental setup and simulation

The initial phase of the experimental setup involves data from two EE sessions with DC experts. Whereas the simulation experiments involve data only from LLMs. Though the sessions with DC experts were group-based [3], the results presented in this article excluded the experts' discussion to ensure a smooth blend with LLMs- generated simulation responses. Figure 1 depicts 4 experimental setups of this study: group-based EE, zero-shot, context, and anchors.

First, with DC experts, the two sessions used the same methodology, tools, and procedure. The EEs study was conducted in Microsoft Teams [35], with Slido for real-time question-and-answer sessions and initial summaries, which were viewed by all members and used by the elicitation facilitator to trigger discussion among experts [56]. Read.AI was used for meeting transcription and recording [46], with the transcription focused more on discussion between experts and out of scope for this paper.

The second to fourth experiments involved the selected models described in Table 2. We prompted model variants using a Python-scripted, stateless Ollama REST API to ensure that prompts from earlier runs do not influence subsequent outputs. Ollama is an open-source platform that supports variant configurations via a model file [41]. To account for variability (i.e., stochastic nature) in generation, multiple runs are commonly used in studies that engage LLMs [2, 5]. We run 50 iterations for each question, both in zero-shot and in context. As for the anchor and list presentation, we run

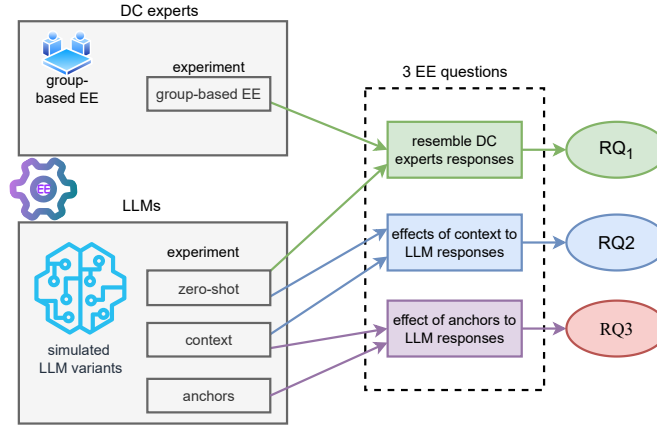


Fig. 1. Experimental setup of DC experts and LLM simulations

10 iterations (i.e., per anchor arm: word, example, low-numeric, and high-numeric), leading to a total of 40 iterations per EE question for comparability with zero-shot and context experiments.

The second experiment is based on zero-shot learning (i.e., no-domain background information on the project, demonstration areas, and DC solutions), as shown in Figure 10. The third experiment on context aligns with the information provided to DC experts, as obtained from the Read.AI transcription. Since LLMs does not work well with Automatic Speech Recognition (ASR) texts [61, 64]. The transcript was post-edited in ChatGPT-4.1, with punctuation restored, grammar corrected, and the text lightly paraphrased into clear text. The resulting paraphrased text was then checked and verified by the authors. The context included detailed background information on the project, demonstration areas, and DC solutions, as shown in Figure 11. Further experiments on anchored prompts (i.e., triggered prompts on statements concerning DC microgrids) are presented in Figure 12.

The analysis compares the experiments:

- The EE with DC experts, and simulated LLMs at zero-shot, answering the first research question, *RQ1*.
- Then, assessment of the effect of context on LLMs responses (i.e., context vs. zero-shot) forms the second research question, *RQ2*.
- Third, to identify the EE judgmental errors by identifying the effect of anchors on LLMs responses (i.e., anchors vs. context), forming the third research question, *RQ3*.
- Furthermore, *RQ1*, *RQ2*, and *RQ3* are all based on three EE questions—feasibility, importance, and barriers.

The *feasibility question* is based on ratings on the overall feasibility of DC solutions. Whereby, the important question was to rate each of the DC solutions. The ratings for the feasibility and importance questions are on a 5-point scale, as indicated in Table 3. Finally, for barrier selection, one must choose the 5 most perceived barriers to the adoption of DC solutions from the 11 listed in Table 4. In addition, the barriers in the figures have been drawn in accordance with the barrier codes listed in the barrier dictionary, Table 4.

3.4 Tests and analytical procedures

3.4.1 Zero-shot vs. DC experts. The comparative assessment of feasibility and importance questions in LLMs using zero-shot versus DC experts' responses involves median values and the 50% span (i.e., Inter-Quartile Range (IQR)) of

Table 4. Barrier Dictionary with Corresponding Codes

Code	Barrier Label
B1	power losses, quality and safety issues
B2	reduced reliability in DC devices
B3	lack of use-cases in which DC is advantageous
B4	uncertain utility interaction (net metering, utility ownership, and agreed standards)
B5	lack of pilot projects
B6	public perception of DC and readiness to 'champion' installations from DC projects
B7	incompatibility of DC systems components
B8	misconception and lack of knowledge lead to lengthy/expensive design and permits
B9	lack of enough trained personnel in DC systems
B10	uncertain regulatory roadmap
B11	high costs of DC solutions

responses across LLMs and the DC experts. Regarding the barrier question, the selection rate was used to assess overlap between the top 5 barriers selected by LLMs and those selected by DC experts.

3.4.2 Context vs. zero-shot, and anchors vs. context. To understand the effect of one condition on the other, for ordinal ratings (i.e., feasibility and importance), we used a Cumulative Link Mixed Model (CLMM). The CLMM is a statistical test for analyzing ordered ordinal data and accounts for random effects between groups [9, 43]. In analyses, one condition (i.e., reference condition) is kept constant, whereby the effect of the second condition (i.e., contrast condition) is expressed relative to the reference condition reported as (*CONTRAST* vs. *REFERENCE*).

The formula for CLMM as in Equation (1):

$$\Pr(Y_i \leq c) = \text{logit}^{-1}(\tau_c - \eta_i - b_p[i] - u_s[i]), \quad c = 0, 1, 2, 3, \quad \tau_0 < \tau_1 < \tau_2 < \tau_3, \quad (1)$$

Where Y_i denotes each variant's feasibility or importance ordinal rating values for each individual observation i . In 5-point ordinal categories 0, 1, 2, 3, 4, maximum cuts are given by $K - 1 = 4$, therefore c is the cut (threshold) index is used in cumulative probabilities, $c = 0, 1, 2, 3$ and a maximum of the τ_c are the cut-points that define the category boundaries are $\tau_0 < \tau_1 < \tau_2 < \tau_3$ estimated in increasing order across ordinal categories.

The η_i is the fixed-effects linear predictor that highlights the relevant coefficient for rating i . Also, the $b_p \sim \mathcal{N}(0, \sigma_p^2)$ and $u_s \sim \mathcal{N}(0, \sigma_s^2)$ are random intercepts per model and per DC solution, respectively. Where, for feasibility, the ($u_s \equiv 0$) as ratings are not per DC solution. As for importance ratings, both random-intercept variables apply, as ratings differ across models and DC solutions.

The estimation of CLMM was performed using the 'R' ordinal package, and results were reported in two ways: odds ratios (ORs) and the probability of higher ratings. The OR for each model is derived from the logit link in Equation (1). At $[OR = 1]$ suggests no effect; $[OR > 1]$ the *CONTRAST* is more likely to increase odds; $[OR < 1]$ more likely the *CONTRAST* condition decreases odds. The probability of higher ratings of each model: $P_{\geq 3} = \Pr(Y \geq 3)$ is derived from fitted thresholds and η_i . It is computed per condition; therefore, a positive Δ indicates that the model is more likely to give higher ratings under the *CONTRAST* condition than the *REFERENCE* condition. Conversely, the chances are lower under the *CONTRAST* condition.

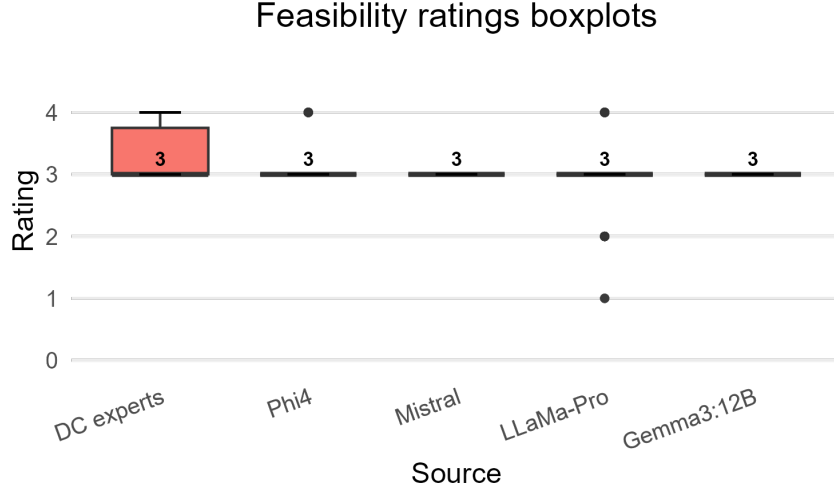


Fig. 2. Feasibility ratings—models vs. DC experts

3.5 Methods for barrier selection

As for the barrier list presentation question, the Jaccard Similarity Index (JSI) was computed. The JSI is the measure of similarity between sets (irrespective of selection order), having a value ranging from 0 to 1 [16, 44]. The closer its value to 1, the more similar the two sets are, and the inverse is true. We computed the JSI for three conditions (i.e., context and zero-shot, context and anchor, and barrier lists) per model.

For two finite sets A and B , the JSI is given by:

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|} \in [0, 1]. \quad (2)$$

4 Results

This section presents the results of the four experiments presented as a comparative assessment, presenting: LLMs at zero-shot versus DC experts, the effect of context relative to zero-shot, and the effect of the anchors relative to context.

4.1 DC experts vs. Models at zero-shot

This part presents results of LLMs at zero-shot with DC experts on three questions: feasibility, importance, and barrier selection.

4.1.1 Feasibility rating question—models vs. DC experts. The results on feasibility ratings are presented in Figure 2, showing DC experts and all models strongly assert on the feasibility of the DC solutions, having [median=3]. All models have a tighter convergence [$IQR = 0$] than a slightly looser one on DC experts [$IQR = 0.75$].

4.1.2 Importance rating question—models vs. DC experts. The results of the importance ratings for each DC solution are summarized in box plots (Figure 3) that compare responses from DC experts and the four LLMs. Overall, the models

captured patterns broadly consistent with DC experts, though differences emerged in the strength of their agreement on specific solutions.

For most DC solutions, the median importance rating across groups was 4 (“very important”), suggesting a shared understanding of the relative significance of these technologies. In nearly all cases, the models exhibited tighter convergence (IQR = 0) than the DC experts, who displayed slightly broader variation (IQR $\approx$ 0.5–1). This indicates that LLM responses were more homogeneous and less dispersed than those from human participants.

Some examples help illustrate these patterns:

- For the DC–DC converter, both Phi4 and Gemma3:12B closely mirrored the experts’ evaluation (median = 4, IQR = 0), showing nearly identical central tendencies and dispersion.
- The DC connectors question revealed similar alignment: Gemma3:12B, Mistral, and Phi4 all matched the experts’ median (4) but were more consistent internally (IQR = 0 vs. experts’ IQR = 1).
- In contrast, the DC measurement device showed slightly wider divergence: LLaMa-Pro had perfect agreement within its own outputs (IQR = 0), while experts showed mild variability (IQR = 0.5), and the other models were more dispersed (IQR = 1).
- For the LVAC–LVDC interlink converter, experts and Gemma3:12B shared the same median rating (4), but the experts’ responses were more cohesive (IQR = 0) than the model’s (IQR = 1), suggesting human participants agreed more strongly on this solution’s importance.

Across all solutions, Phi-4 most frequently reproduced the experts’ median ratings (10 of 11 solutions), followed by Gemma3:12B (7 solutions), Mistral (4), and LLaMa-Pro (1). When considering both medians and IQRs together, Phi4 and Gemma3:12B consistently emerged as the closest approximations to expert judgment, combining accurate central tendencies with stable consensus across multiple DC solutions.

In summary, while no model perfectly replicated expert responses across all criteria, Phi4 and Gemma3:12B demonstrated the strongest alignment, suggesting they capture the general structure of expert reasoning more reliably than the other models.

4.1.3 Barrier selection question—models vs. DC experts. Figure 4 presents the barrier selection rate in two facets. First DC experts with pooled-LLMs (i.e., combined responses from all models), then per model. Considering the DC experts with pooled-LLMs: indicate LLMs highly selected 1st to 5th barrier, contrary to DC experts who highly selected B6, B8, B9, B10, and B11. Per model comparison relative to DC experts’ highly rated barriers, shows: Gemma3:12B also highly rated 3 – B8, B9, and B11; Phi4 highly rated 1 – B10; none from Mistral or LLaMa-Pro.

4.2 Context vs. zero-shot

This section presents results on the LLMs effect under context condition relative to the zero-shot condition for three EE questions (i.e., feasibility, importance, and barriers) for each model.

4.2.1 Feasibility question—context vs zero-shot. Table 5 shows the effect of context compared to zero-shot for the feasibility question. Three models (i.e., Phi4, Gemma3:12B, and Mistral) had no effect, having [(OR=1), at 95%(0.852, 1.174)]. On the other hand, (i.e., LLaMa-Pro) shows insignificant increasing odds at context relative to zero-shot [(OR=1.104), at 95%CI(0.927, 1.314)]. Furthermore, Figure 5 shows there are no changes in probabilities of higher ratings for Gemma3:12B, and Mistral between context and zero-shot conditions. As for Phi4, the likelihood of higher ratings is

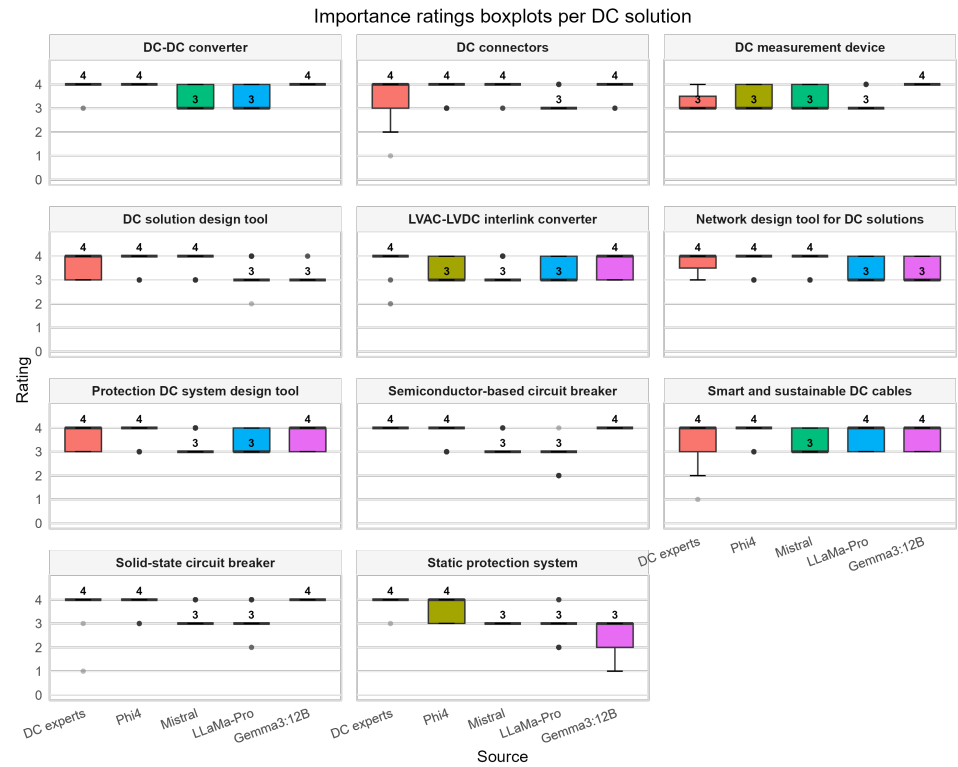


Fig. 3. Importance ratings—models vs. DC experts

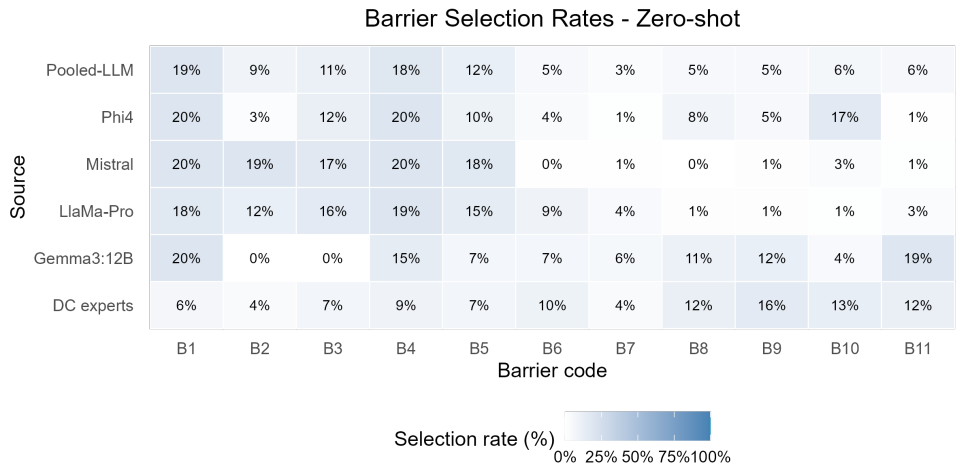
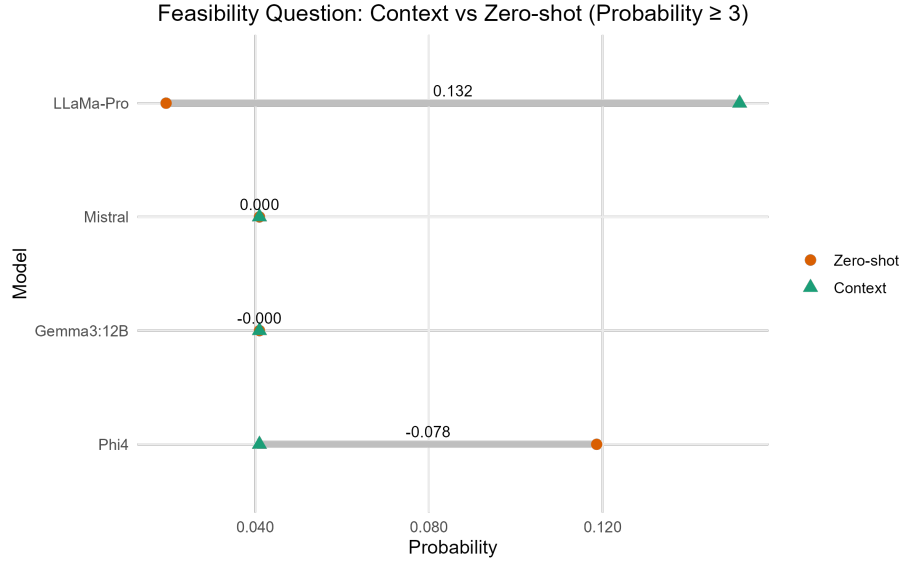


Fig. 4. Barrier selection—models vs. DC experts

Table 5. Feasibility ratings—context vs. zero-shot

Model	OR	CI_l95	CI_u95
Gemma3:12B	1.000	0.852	1.174
LlaMa-Pro	1.104	0.927	1.314
Mistral	1.000	0.852	1.174
Phi4	1.000	0.852	1.174

Fig. 5. Feasibility question - the Δ in probabilities of higher ratings.

The annotations are the Δ between each model's $P \geq 3$ at context and zero-shot. For instance, Mistral had the same value at both conditions, hence $\Delta = 0.000$.

high under zero-shot relative to context [$\Delta = -0.078$], while for LLaMa-Pro, the probability increases in context relative to zero-shot [$\Delta = 0.132$].

4.2.2 Importance question—context vs zero-shot. This part also reports the effect in two aspects: the OR per model in Table 6, and the probability of higher ratings indicated by the Δ in probabilities per model per DC solution in Figure 6.

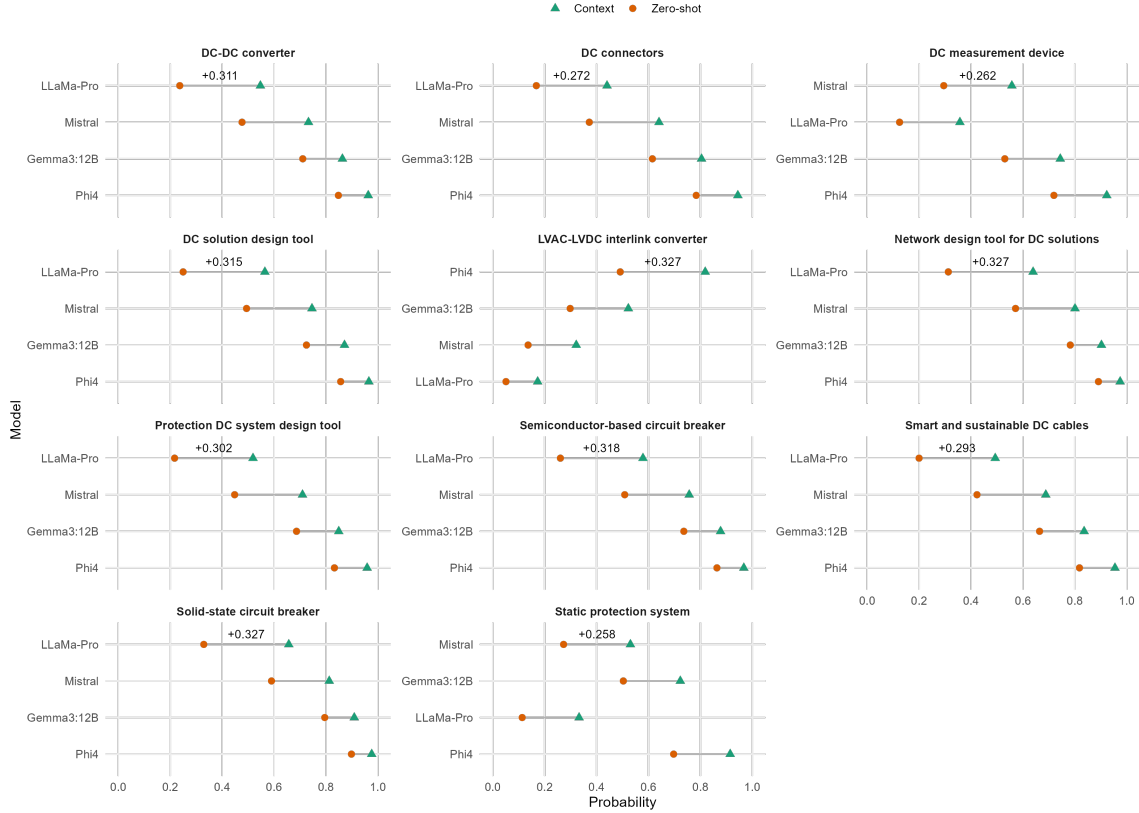
The OR results, as in Table 6, indicate all models have a significant increase in OR under the context condition relative to the zero-shot condition: Phi4 (OR=4.675, 95%CI[3.680, 5.940]), LLaMa-Pro (OR=3.898, 95%CI[3.194, 4.757]), Mistral (OR=3.010, 95%CI[2.474, 3.661]), and Gemma3:12B (OR=2.574, 95%CI[2.287, 2.895]).

Overall, Figure 6 shows that none of the models have a negative Δ in probabilities, which indicates a positive shift towards higher ratings in context compared to zero-shot for all models.

LLaMa-Pro leads other models in having the highest Δ value in 8 DC solutions. This implies that it has a higher chance of achieving higher ratings across the 8 DC solutions in context than with zero-shot. The DC solutions that LLaMa-Pro lead are: DC-DC converters, DC connectors, DC solution design tool, network design tool for DC solutions,

Table 6. Importance ratings—context vs. zero-shot

Model	OR	CI_l95	CI_u95
Gemma3:12B	2.574	2.287	2.895
LlaMa-Pro	3.898	3.194	4.757
Mistral	3.010	2.474	3.661
Phi4	4.675	3.680	5.940

Importance Question: Context vs Zero-shot (Probability ≥ 3)Fig. 6. Importance question - the Δ in probabilities of higher ratings.

The annotations in each sub-figure representing the DC solutions are only for a model with the highest Δ of probabilities between context and zero-shot conditions.

protection DC system design tool, semiconductor-based circuit breaker, smart and sustainable DC cables, and solid-state circuit breaker. Mistral leads in two DC solutions: the DC measurement device and static protection system. Phi4 leads to one DC solution: the LVAC-LVDC interlink converter, and Gemma3:12B to none.

4.2.3 Barrier selection question—context vs zero-shot. Table 7 presents the JSI values and Figure 7 indicates a change in model preferences on the top-5 barriers between the conditions.

Table 7. JSI on top-5 barriers—context vs. zero-shot

Model	$JSI(\text{Context}, \text{Zero-shot})$
Gemma3:12B	0.667
LLaMa-Pro	1.000
Mistral	1.000
Phi4	0.429

The JSI was computed for the 5 barriers with the highest selection rate in each model, across the context and zero-shot conditions.

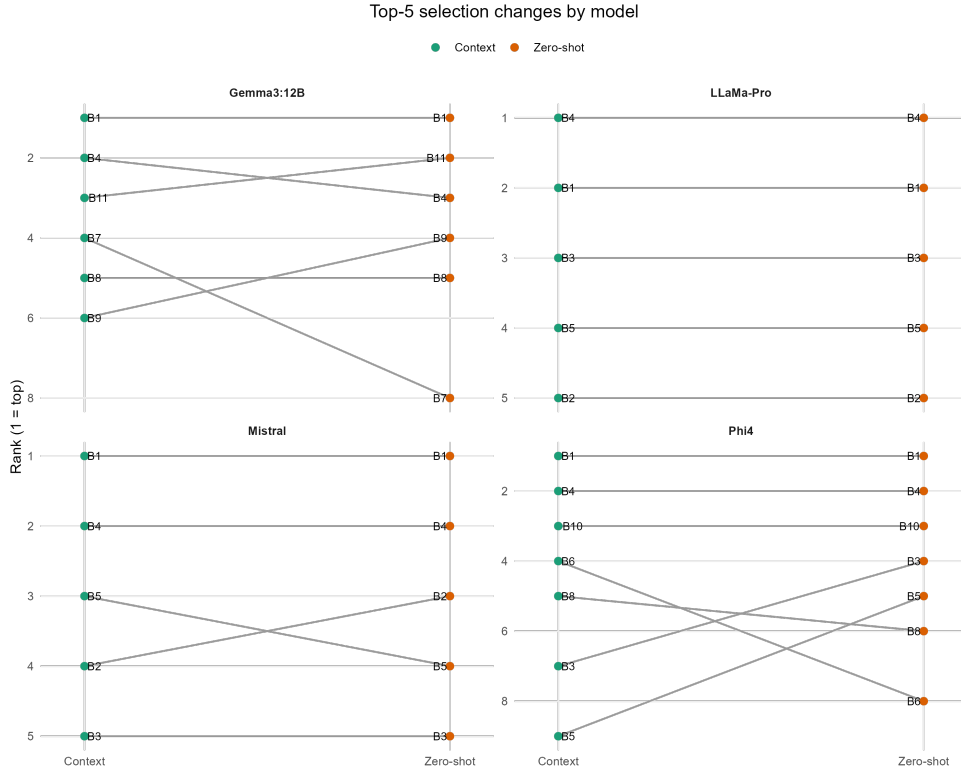


Fig. 7. Barrier selection—context vs zero-shot

This figure captures the changes in barrier selection pattern for each model under the context relative to the zero-shot.

LLaMa-Pro and Mistral selected the same barriers under the context condition relative to zero-shot, [$JSI=1.000$]. The selection pattern is captured in Figure 7, which shows that LLaMa-Pro maintained the same barrier positions, while Mistral flipped third and fourth positions between the same barriers (i.e., B2 and B5). Furthermore, there were observed changes in the top-5 barrier selection behavior on Gemma3:12B [$JSI=0.667$], and highest on Phi4 [$JSI=0.429$].

Table 8. Feasibility ratings—anchors vs. context

Model	Arm	OR	CI_l95	CI_u95
LlaMa-Pro	Word	3.729	2.005	6.933
LlaMa-Pro	Example	2.844	1.479	5.468
LlaMa-Pro	Low numeric	2.343	1.202	4.565
LlaMa-Pro	High numeric	4.118	2.215	7.654
Mistral	Word	1.359	0.412	4.489

4.3 Anchors vs. context

The results presented in this section assess the availability of judgmental errors when prompts are anchored on four arms (i.e., word, example, low-numeric, and high-numeric) on three EE questions as well. Three judgmental errors are examined in the results: overconfidence, over-extremity, and anchoring and adjustment.

4.3.1 Feasibility rating question—anchors vs. context. Table 8 shows that two models significantly altered their responses per anchor arm relative to the context. All other models (i.e., Phi4, Gemma3:12B) had no effect in all arms relative to the context condition, $[OR = 1]$, therefore were omitted from the table. LlaMa-Pro adjusted in each of the anchor arms; word, example, low numeric, and high numeric had significant increasing odds: $[(OR=3.729), 95\% CI(2.005, 6.933)]$, $[(OR=2.844), 95\% CI(1.479, 5.468)]$, $[(OR=2.343), 95\% CI(1.202, 4.565)]$, and $[(OR=4.118), 95\% CI(2.215, 7.654)]$ respectively. The observed rhythmic tone has a low effect on low-numeric and a higher effect on high-numeric, exhibiting the *anchoring and adjustment* judgmental error. In addition, though at an insignificant effect, Mistral also adjusted to the word anchor $[(OR=1.359), 95\% CI(0.412, 4.489)]$. The effect, which also positively increases its Δ on probability of higher ratings $[\Delta = 0.007]$, even more than LlaMa-Pro's $[\Delta = 0.003]$ at the word anchor as presented in Figure 8.

On the other hand, Gemma3:12B and Phi4 exhibited *overconfidence* and *over-extremity* behaviors by not adjusting their ratings under any of the anchor arms relative to context, but also being too rigid, both having $[\Delta = 0]$ in each of the anchor arms.

4.3.2 Importance rating question—anchors vs. context. Table 9 presents the effect per model for the importance question. As for LlaMa-Pro, the effect, though it is significant at high-numeric and not significant at low-numeric, anchoring and adjustments judgmental error is observed as in the feasibility question: $[(OR=1.857), 95\% CI(1.468, 2.349)]$ at high-numeric, and $[(OR=1.202), 95\% CI(0.952, 1.518)]$ at low-numeric. In addition, Mistral and Phi4 behaved as LlaMa-Pro, though not with significant effect.

On the other hand, for the word anchor Phi4, Gemma3:12B, and LlaMa-Pro had significantly higher increasing odds of: $[(OR=10.681), 95\% CI(4.135, 27.590)]$, $[(OR=1.721), 95\% CI(1.190, 2.489)]$, and $[(OR=1.393), 95\% CI(1.103, 1.759)]$ respectively. Further analysis is on the probabilities of higher ratings for each model in each anchor relative to the context condition. Since in both conditions (i.e., context and each anchor arm) there are no ratings less than 3, as presented in Figure 9, therefore, by the trivial, its value is 1.

4.3.3 Barrier selection question—anchors vs context. Table 10 indicates the JSI in two aspects. First context with pooled anchors (i.e., treat ratings irrespective of anchor arm), then per anchor arm. Results indicate that at pooled anchors, there is no difference in how three models (i.e., Phi4, Mistral, and Gemma3:12B) selected the top 5 barriers, hence $[JSI = 1]$. Only LlaMa-Pro altered its selection pattern, $[JSI = 0.667]$.

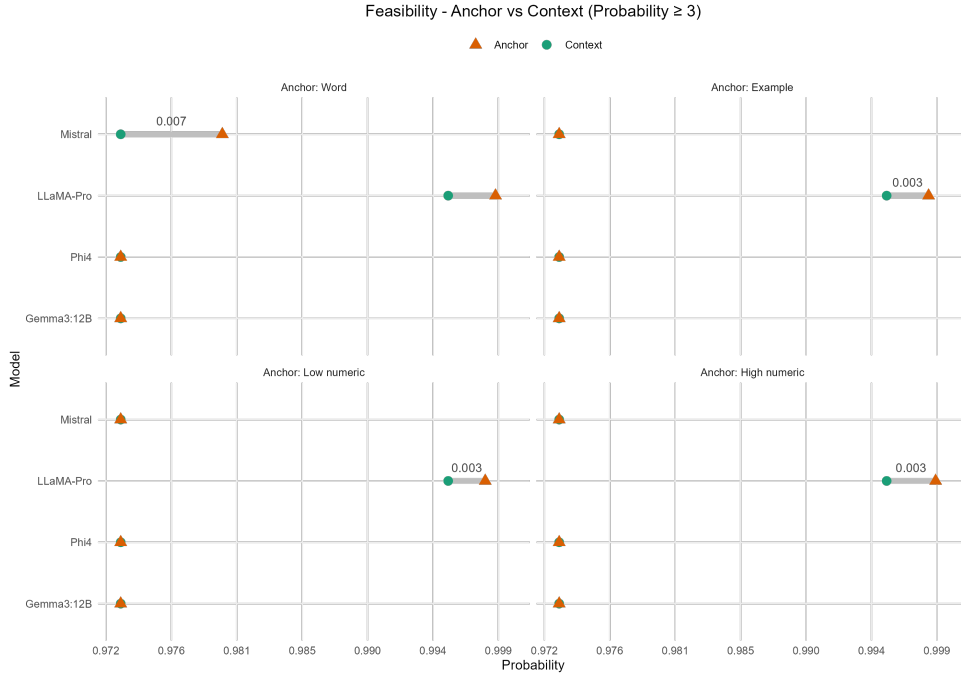
Fig. 8. Feasibility question— Δ in probabilities of higher ratings

Table 9. Importance ratings—anchors vs. context

Model	Arm	OR	CI_l95	CI_u95
Gemma3:12B	Word	1.721	1.190	2.489
LlaMa-Pro	Word	1.393	1.103	1.759
Phi4	Word	10.681	4.135	27.590
LlaMa-Pro	High numeric	1.857	1.468	2.349
Mistral	High numeric	0.584	0.446	0.764
Phi4	High numeric	1.322	0.815	2.143
LlaMa-Pro	Low numeric	1.202	0.952	1.518
Mistral	Low numeric	0.118	0.089	0.157
Phi4	Low numeric	0.369	0.253	0.540

Per model, per anchor arm; Gemma3:12B has $[JSI = 1]$ at the context condition for each of the three arms (i.e., example, low-numeric, and high-numeric). All other models have a perfect index $[JSI = 1]$ in two different anchor arms on each: LLaMa-Pro on word and low-numeric, Mistral on low-numeric and high-numeric, and Phi4 on word and high-numeric. All other remaining arms, for each model, have $[JSI = 0.667]$.

This implies that each model mostly maintained its selection of the top 5 barriers across anchor arms, relative to context. This implies the models showed little to no variation at anchors relative to context; therefore, they have exhibited overconfidence and over-extremity in their selection behavior.

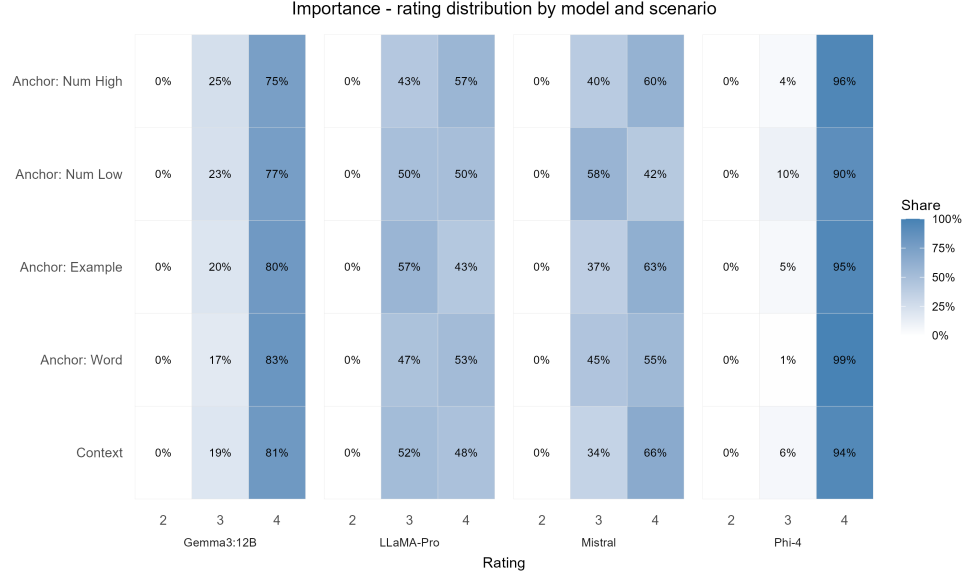


Fig. 9. Importance rating distribution—anchors vs. context

Table 10. JSI on top-5 barriers—anchors vs. context

Model	$JSI(\text{Ctx, Anchors})$	$JSI(\text{Ctx, Word})$	$JSI(\text{Ctx, Example})$	$JSI(\text{Ctx, Low-num})$	$JSI(\text{Ctx, High-num})$
Gemma3:12B	1.000	0.667	1.000	1.000	1.000
LLaMa-Pro	0.667	1.000	0.667	1.000	0.667
Mistral	1.000	0.667	0.667	1.000	1.000
Phi4	1.000	1.000	0.667	0.667	1.000

5 Discussion

This section maps our findings from Section 4 to the research questions and related literature. Finally, some implications for the design of EE systems that collaborate with Human-LLMs are also provided.

5.1 Answers to Research Questions

5.1.1 Models with DC experts. The experimental analyses of models at zero-shot with the DC experts aimed to answer the first research question (i.e., *RQ1*). The question aims to understand the capabilities of LLMs to emulate experts. Based on the results obtained and presented at Section 4.1, it is reasonable to say that it is possible to emulate experts with LLMs. However, several aspects should be considered, for instance, the varied behavior of model variants towards questions with similar rating scales, such as ordinal ratings for feasibility and importance questions.

5.1.2 Effect under context relative to zero-shot. The experimental assessment on context with zero-shot targets to answer the second research question (i.e., *RQ2*). It aims to understand the effect of context on LLMs performance in EE questions. The obtained results, presented in Section 4.2, show that there is an effect on LLMs responses in the context condition relative to the zero-shot, although the effect is not consistent across all questions and models. While some models, like LLaMa-Pro, affected feasibility and importance questions, others, like Gemma3:12B, Phi4, and Mistral,

only positively affected the importance question. On barrier selection patterns between the conditions, LLaMa-Pro and Mistral were rigid in selection patterns in the two conditions, while Phi4 and Gemma3:12B adjusted. This implies that providing LLMs with more specific context related to the subject under study can positively enhance their ratings.

5.1.3 Judgmental errors under anchors relative to context. The analyses of the anchors and context experiments focus on answering the third research question (i.e., *RQ3*). The question aims to identify the prevalence of judgment errors when EE audiences are LLMs rather than human experts. Our experiments with models reveal the likelihood of judgmental errors, which are reported to be possible when EE involves human experts. As in other experiments reported in this study, no single model exhibited all three judgmental errors (i.e., overconfidence, over-extremity, and anchoring and adjustment). Our results highlight the need for caution in the use of anchors when dealing with LLMs for EE studies.

5.2 Positioning Within Related Work

5.2.1 Under-specificity in EE. In EE, under-specificity happens when the given information is not specific enough to provide a clear direction to experts on how to approach EE questions [25, 36, 47]. In our experimental settings, this could be related to a zero-shot experiment. Contrary to EE involving experts, even at under-specificity models were able to emulate experts' responses. This implies that, even with minimal specificity, LLMs performs better than experts when given well-structured prompts. In addition, under-specificity has been shown to have minimal impact when the elicitation audience is LLMs rather than experts.

5.2.2 Specificity in EE. Specificity, as in EE, implies given the required specific information that could keep experts more informed and support eliciting useful judgments [25]. As for this study, it could be derived from the context experimental setup in Section 4.2. As for the EE involving experts, this is expected to enhance the quality of gathered responses [25]. Conversely, our findings differed across models and the EE question. Relating our findings to studies in LLMs [30]; there are concerns of observed effects being triggered by primacy, lost-in-the-middle, or recency effects.

5.2.3 Primacy, lost-in-the-middle, and recency effect. Literature relates LLMs the behavior of highly considering the first part in the list (i.e., primacy effect) to their not adjusting to information. Because they tend to bypass some parts of the information in the middle (i.e., lost-in-the-middle) and consider the last items in the information (i.e., recency effect) [30]. In our study, at Section 4.2.1 and Section 4.3.1, the LLMs that did not alter their feasibility ratings in response to changes in context are Gemma3:12B and Mistral. However, in assessing the primacy effect using selection rates at Section 4.1.3, Mistral showed a stronger primacy effect than Gemma3:12B, with higher selection rates for the first 5 barriers. The most intriguing observation concerns the last barriers in the list (i.e., 8th to 11th), where Mistral has very low rates compared to the higher scores shown by Gemma3:12B. This implies that though rigidity in feasibility ratings observed in Gemma3:12B might be linked to the challenge of the recency effect, the same cannot be derived for Mistral, which also provided rigid scores.

5.3 Implications for Human-LLM Expert Elicitation

Our findings have several implications for the design and study of asynchronous expert elicitation systems. First, the ability of open-source LLMs to emulate expert judgments and to exhibit similar cognitive biases highlights their potential role as co-participants in structured expert elicitation rather than mere computational tools. This opens a new research direction on AI-assisted expert elicitation, where models can complement human expertise asynchronously and reduce the logistical burdens of traditional group sessions.

Second, the observed sensitivity of models to context and anchors suggests that interaction design choices—such as prompt framing, information ordering, and transparency of anchors—directly influence model behavior. This calls for the development of elicitation interfaces that make these mechanisms visible and controllable by moderators. In addition, designs that consider weighting each *entity* in the pool of LLMs with experts in the EE are vital, given the exhibited variability in LLMs’s responses. We advice the weight be considered per EE question or phase, and not per session given the exhibited variability in LLMs’ responses. Finally, our results motivate further investigation into adaptive elicitation platforms, where human and LLM participants jointly refine prompts, validate reasoning chains, share evidence, and calibrate uncertainty, bridging human judgment and machine intelligence within a single interactive framework.

5.4 Generalizability

Regarding generalizability, we position the findings from this work as applicable primarily to rating-based structured judgment tasks. Though we expect to orchestrate the LLMs in group elicitation formats (e.g., facilitated workshops, Delphi-like rounds of discussion with controlled feedback, or even deliberative expert panels), as for these results, we did not model the interaction between LLMs. Generalization across domains and model families depends on the specific prompt design and model versions used, as small changes can alter outputs. Replication across additional domains, broader expert samples, and truly interactive elicitation protocols is therefore required to establish robustness and boundary conditions.

6 Conclusion, Limitations, and Future Works

This section presents the study’s conclusion, the observed limitations arising from the experimental setup, and the results. The section also concludes with a possible future extension to this work.

6.1 Conclusion

By achieving the goal of engaging LLMs in structured EE through three research questions, answered by four experiments, our setup explores and compares the capabilities of four open-source LLMs on simulated structured EE questions.

To answer the first research question, we started with minimal information (i.e., zero-shot) and found that LLMs performed better than DC experts. The second research question prompts were enriched with more context, as those given to DC experts, and we found that there are increasing odds of LLMs with context relative to zero-shot, though not consistently across models. Furthermore, for the third research question, we provided four context prompts to assess judgmental errors in EE involving experts, and we found that overconfidence, over-extremity, and anchoring and adjustment were captured based on LLMs performance at anchors relative to the context condition.

6.2 Limitations

Despite the objectives of this research having been achieved, we acknowledge certain limitations that should be considered when interpreting the results.

Though the sample of DC experts explored in this research work resembles that of several other EE studies [18, 32, 67, 68], it limits the exploration of quantitative comparisons with LLMs responses. Hence, the comparison of DC experts with LLMs in the *RQ1* relied on medians and IQRs through the box plots for feasibility and importance questions.

Also, our contribution is inconclusive if we were to choose which model would better fit EE given those engaged in the experiments herein. Although for each of the experimental designs presented in this study, there are leading models, performance differs across EE question types despite alterations in the level of prompt enrichment (i.e., from zero-shot